



Analisi e Progetto di Algoritmi

Laurea Triennale in Informatica
(codice: E3101Q113)

AA 2025/2026

Longest Common Subsequence
(calcolo del valore ottimo e ricostruzione della soluzione)

Docente: Raffaella Rizzi

[Concetti di base]

Sequenza

$$X = \langle 2, 4, 10, 5, 9, 11 \rangle$$

[Concetti di base]

Sequenza

$$X = \langle 2, 4, 10, 5, 9, 11 \rangle$$

Prefisso di una sequenza

$$X_3 = \langle 2, 4, 10 \rangle$$

$$X_6 = \langle 2, 4, 10, 5, 9, 11 \rangle$$

$$X_0 = \langle \rangle$$

[Concetti di base]

Sequenza

$$X = \langle 2, 4, 10, 5, 9, 11 \rangle$$

Sottosequenza

$$\langle 4, 5, 9 \rangle$$

$$\langle \rangle$$

$$\langle 4, 9, 5 \rangle$$

[Concetti di base]

Sottosequenza comune di due sequenze X e Y

$X = \langle 1, 13, 5, 3, 1, 12, 8, 11, 6, 10, 10 \rangle$

$Y = \langle 1, 5, 5, 2, 3, 1, 12, 8, 8, 10 \rangle$

[Concetti di base]

Sottosequenza comune di due sequenze X e Y

$X = \langle 1, 13, 5, 3, 1, 12, 8, 11, 6, 10, 10 \rangle$

$Y = \langle 1, 5, 5, 2, 3, 1, 12, 8, 8, 10 \rangle$

$\langle 5, 3, 1, 8, 10 \rangle$

[Problema LCS(X,Y)]

Date due sequenze

$$X = \langle x_1, x_2, \dots, x_m \rangle, Y = \langle y_1, y_2, \dots, y_n \rangle$$

trovare la più lunga sottosequenza comune di X e Y

$(m,n) \rightarrow$ dimensione del problema

[Come si risolve il problema?]

- Scompongo in sottoproblemi
- Individuo la soluzione di ogni sottoproblema tramite un'equazione di ricorrenza
 - I. Passo ricorsivo
 - II. Casi base

[Sottoproblema]

Sottoproblema di dimensione (i,j)

[Sottoproblema]

Sottoproblema di dimensione (i,j)

→ Trovare $\text{LCS}(X_i, Y_j)$

$i \in \{0, 1, \dots, m\}$

$j \in \{0, 1, \dots, n\}$

[Sottoproblema]

Sottoproblema di dimensione (i,j)

→ Trovare $\text{LCS}(X_i, Y_j)$

$i \in \{0, 1, \dots, m\}$

$j \in \{0, 1, \dots, n\}$

Numero totale di sottoproblemi: $(m+1)(n+1)$

[Sottoproblema]

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$i \in \{0, 1, \dots, m\}$

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Numero totale di sottoproblemi: $(m+1)(n+1)$

$$c_{i,j} = |\text{LCS}(X_i, Y_j)|$$

[Sottoproblema]

Sottoproblema di dimensione (i,j)

→ Trovare $\text{LCS}(X_i, Y_j)$

$i \in \{0, 1, \dots, m\}$

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Numero totale di sottoproblemi: $(m+1)(n+1)$

$$c_{m,n} = |\text{LCS}(X_m, Y_n)| = |\text{LCS}(X, Y)|$$

[Equazioni di ricorrenza]

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$i = 0 \vee j = 0$ (CASI BASE)

$$\text{LCS}(X_i, Y_j) =$$

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$$\text{LCS}(X_i, Y_j) = \langle \rangle$$

$$X_i = \langle x_1, x_2, \dots, x_{i-1}, x_i \rangle$$

$$Y_j = \langle y_1, y_2, \dots, y_{j-1}, y_j \rangle$$

$i > 0 \wedge j > 0$ (PASSO RICORSIVO)

$$x_i = y_j \Rightarrow \text{LCS}(X_i, Y_j) =$$

$$x_i \neq y_j \Rightarrow \text{LCS}(X_i, Y_j) =$$

[Equazioni di ricorrenza]

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$$x_i = y_j \Rightarrow \text{LCS}(X_i, Y_j) = \text{LCS}(X_{i-1}, Y_{j-1}) + \langle x_i \rangle$$

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$$x_i \neq y_j \Rightarrow \text{LCS}(X_i, Y_j) = \max\{\text{LCS}(X_{i-1}, Y_j), \text{LCS}(X_i, Y_{j-1})\}$$

[Equazioni di ricorrenza]

$i = 0 \vee j = 0$ (CASI BASE)

$$\text{LCS}(X_i, Y_j) = \langle \rangle \Rightarrow c_{i,j} =$$

$$X_i = \langle x_1, x_2, \dots, x_{i-1}, x_i \rangle$$

$$Y_j = \langle y_1, y_2, \dots, y_{j-1}, y_j \rangle$$

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$$Y_j = \langle y_1, y_2, \dots, y_{j-1}, y_j \rangle$$

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$$x_i = y_j \Rightarrow \text{LCS}(X_i, Y_j) = \text{LCS}(X_{i-1}, Y_{j-1}) + \langle x_i \rangle$$

$$\Rightarrow c_{i,j} = c_{i-1,j-1} + 1$$

$$x_i \neq y_j \Rightarrow \text{LCS}(X_i, Y_j) = \max\{\text{LCS}(X_{i-1}, Y_j), \text{LCS}(X_i, Y_{j-1})\}$$

$$\Rightarrow c_{i,j} =$$

[Equazioni di ricorrenza]

$i = 0 \vee j = 0$ (CASI BASE)

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$$X_i = \langle x_1, x_2, \dots, x_{i-1}, x_i \rangle$$

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$$\Rightarrow c_{i,j} = \max\{c_{i-1,j}, c_{i,j-1}\}$$

[Equazioni di ricorrenza]

$i = 0 \vee j = 0$ (CASI BASE)

$$c_{i,j} = 0$$

$$X_i = \langle x_1, x_2, \dots, x_{i-1}, x_i \rangle$$

$$Y_j = \langle y_1, y_2, \dots, y_{j-1}, y_j \rangle$$

$i > 0 \wedge j > 0$ (PASSO RICORSIVO)

$$x_i = y_j \Rightarrow c_{i,j} = c_{i-1,j-1} + 1$$

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[Soluzione del problema (DP)]

1. Calcolo del valore ottimo $c_{m,n}$ con un algoritmo bottom-up
 - memorizzazione dei coefficienti
2. Ricostruzione di $\text{LCS}(X, Y)$

[Algoritmo bottom-up]

C						
	y_1	...	y_j	...	y_n	
						0
x_1						1
...						...
x_i						i
...						...
x_m						m
	0	1	...	j	...	n

[Algoritmo bottom-up]

C							
		y_1	...	y_j	...	y_n	
							0
x_1							1
...							...
x_i				$c_{i,j}$			i
...							...
x_m							m
	0	1	...	j	...	n	

[Algoritmo bottom-up]

C							
		y_1	...	y_j	...	y_n	
							0
x_1							1
...							...
x_i				$c_{i,j}$			i
...							...
x_m						$c_{m,n}$	m
	0	1	...	j	...	n	

[Algoritmo bottom-up]

C							
		y_1	...	y_j	...	y_n	
		0	0	0	0	0	0
x_1	0						1
...	0						...
x_i	0			$c_{i,j}$			i
...	0						...
x_m	0					$c_{m,n}$	m
	0	1	...	j	...	n	

[Algoritmo bottom-up]

C							
		y_1	...	y_j	...	y_n	
	0	0	0	0	0	0	0
x_1	0						1
...	0						...
x_i	0			$c_{i,j}$			i
...	0						...
x_m	0					$c_{m,n}$	m
	0	1	...	j	...	n	

if $x_i = y_j$ **then**
 $c_{i,j} = c_{i-1,j-1} + 1$
else
 $c_{i,j} = \max\{c_{i-1,j}, c_{i,j-1}\}$

[Algoritmo bottom-up]

C						
		y_1	...	y_j	...	y_n
	0	0	0	0	0	0
x_1	0					
...	0					
x_i	0			$c_{i,j}$		
...	0					
x_m	0					$c_{m,n}$
	0	1	...	j	...	n

if $x_i = y_j$ **then**
 $c_{i,j} = c_{i-1,j-1} + 1$
else
 $c_{i,j} = \max\{c_{i-1,j}, c_{i,j-1}\}$

[Algoritmo bottom-up]

C						
		y_1	...	y_j	...	y_n
	0	0	0	0	0	0
x_1	0					
...	0					
x_i	0					
...	0					
x_m	0					
	0	1	...	j	...	n

if $x_i = y_j$ then

$$C_{i,j} = C_{i-1,j-1} + 1$$

else

$$C_{i,j} = \max\{C_{i-1,j}, C_{i,j-1}\}$$

[Esercizio di riempimento di C]

C		2	5	12	2	3	12	1	30	
										0
2										1
10										2
5										3
3										4
1										5
12										6
	0	1	2	3	4	5	6	7	8	

$X = \langle 2, 10, 5, 3, 1, 12 \rangle$

$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$

[Esercizio di riempimento di C]

C		2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0	0
2	0									1
10	0									2
5	0									3
3	0									4
1	0									5
12	0									6
	0	1	2	3	4	5	6	7	8	

$X = \langle 2, 10, 5, 3, 1, 12 \rangle$

$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$

[Esercizio di riempimento di C]

$$C_{1,1} = ?$$

C		2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0	0
2	0	?								1
10	0									2
5	0									3
3	0									4
1	0									5
12	0									6
	0	1	2	3	4	5	6	7	8	

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

Esercizio di riempimento di C

$$C_{1,1} = C_{0,0} + 1$$

C		2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0	0
2	0	1								1
10	0									2
5	0									3
3	0									4
1	0									5
12	0									6
	0	1	2	3	4	5	6	7	8	

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

[Esercizio di riempimento di C]

$$C_{1,2} = ?$$

C		2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0	0
2	0	1	?							1
10	0									2
5	0									3
3	0									4
1	0									5
12	0									6
	0	1	2	3	4	5	6	7	8	

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

[Esercizio di riempimento di C]

$$c_{1,2} = \max(c_{0,2}, c_{1,1})$$

C		2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0	0
2	0	1	1							1
10	0									2
5	0									3
3	0									4
1	0									5
12	0									6
	0	1	2	3	4	5	6	7	8	

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

[Esercizio di riempimento di C]

$$C_{1,3} = ?$$

C		2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0	0
2	0	1	1	?						1
10	0									2
5	0									3
3	0									4
1	0									5
12	0									6
	0	1	2	3	4	5	6	7	8	

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

[Esercizio di riempimento di C]

$$c_{1,3} = \max(c_{0,3}, c_{1,2})$$

C		2	5	12	2	3	12	1	30	
		0	0	0	0	0	0	0	0	0
2	0	1	1	1						1
10	0									2
5	0									3
3	0									4
1	0									5
12	0									6
	0	1	2	3	4	5	6	7	8	

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

[Esercizio di riempimento di C]

$$C_{1,4} = ?$$

C		2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0	0
2	0	1	1	1	?					1
10	0									2
5	0									3
3	0									4
1	0									5
12	0									6
	0	1	2	3	4	5	6	7	8	

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

Esercizio di riempimento di C

$$C_{1,4} = C_{0,3} + 1$$

C		2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1					1
10	0									2
5	0									3
3	0									4
1	0									5
12	0									6
	0	1	2	3	4	5	6	7	8	

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

[Esercizio di riempimento di C]

$$c_{1,5} = \max(c_{0,5}, c_{1,4})$$

C		2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1				1
10	0									2
5	0									3
3	0									4
1	0									5
12	0									6
	0	1	2	3	4	5	6	7	8	

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

[Esercizio di riempimento di C]

$$c_{1,6} = \max(c_{0,6}, c_{1,5})$$

C		2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1			1
10	0									2
5	0									3
3	0									4
1	0									5
12	0									6
	0	1	2	3	4	5	6	7	8	

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

[Esercizio di riempimento di C]

$$c_{1,7} = \max(c_{0,7}, c_{1,6})$$

C		2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1		1
10	0									2
5	0									3
3	0									4
1	0									5
12	0									6
	0	1	2	3	4	5	6	7	8	

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

[Esercizio di riempimento di C]

$$c_{1,7} = \max(c_{0,7}, c_{1,6})$$

C		2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1	1
10	0									2
5	0									3
3	0									4
1	0									5
12	0									6
	0	1	2	3	4	5	6	7	8	

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

[Esercizio di riempimento di C]

C		2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	1	2
5	0	1	2	2	2	2	2	2	2	3
3	0	1	2	2	2	3	3	3	3	4
1	0	1	2	2	2	3	3	4	4	5
12	0	1	2	3	3	3	4	4	4	6
	0	1	2	3	4	5	6	7	8	

$X = \langle 2, 10, 5, 3, 1, 12 \rangle$

$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$

[Algoritmo bottom-up]

```
int calcola_ottimo(X,Y)
  C ← matrice di m+1 righe e n+1 colonne
  for i from 0 to m do
    C[i,0] ← 0
  for j from 0 to n do
    C[0,j] ← 0
  for i from 1 to m do
    for j from 1 to n do
      if  $x_i = y_j$  then
        C[i,j] ← C[i-1,j-1] + 1
      else
        C[i,j] ← max{C[i-1,j], C[i,j-1]}
  return C[m,n]
```


[Soluzione del problema (DP)]

1. Calcolo del valore ottimo $c_{m,n}$ con un algoritmo bottom-up
 - memorizzazione dei coefficienti
2. Ricostruzione di $\text{LCS}(X, Y)$

[Ricostruzione di $\text{LCS}(X_i, Y_j)$]

C							
		y_1	...	y_j	...	y_n	
x_1 ... x_i ... x_m		0	0	0	0	0	0
		0					1
		0					...
		0					i
		0					...
		0					m
		0	1	...	j	...	n

[Ricostruzione di $LCS(X_i, Y_j)$

C							
		y_1	...	y_j	...	y_n	
		0	0	0	0	0	0
x_1	0						1
...	0						
x_i	0						
...	0						
x_m	0						
		0	1	...	j		

if $x_i = y_j$ then

[Ricostruzione di $\text{LCS}(X_i, Y_j)$

C						
		y_1	...	y_j	...	y_n
		0	0	0	0	0
x_1	0					
...	0					
x_i	0					
...	0					
x_m	0					
		0	1	...	j	...

if $x_i = y_j$ then
 $\text{LCS}(X_i, Y_j) = \text{LCS}(X_{i-1}, Y_{j-1}) + \langle x_i \rangle$

[Ricostruzione di $\text{LCS}(X_i, Y_j)$

C						
		y_1	...	y_j	...	y_n
		0	0	0	0	0
x_1	0					
...	0					
x_i	0					
...	0					
x_m	0					
		0	1	...	j	

if $x_i = y_j$ **then**
 $\text{LCS}(X_i, Y_j) = \text{LCS}(X_{i-1}, Y_{j-1}) + \langle x_i \rangle$
else
if $C[i, j] = C[i-1, j]$ **then**

[Ricostruzione di $\text{LCS}(X_i, Y_j)$]

C						
		y_1	...	y_j	...	y_n
		0	0	0	0	0
x_1	0					
...	0					
x_i	0					
...	0					
x_m	0					
		0	1	...	j	

if $x_i = y_j$ **then**
 $\text{LCS}(X_i, Y_j) = \text{LCS}(X_{i-1}, Y_{j-1}) + \langle x_i \rangle$
else
if $C[i, j] = C[i-1, j]$ **then**
 $\text{LCS}(X_i, Y_j) = \text{LCS}(X_{i-1}, Y_j)$

[Ricostruzione di $LCS(X_i, Y_j)$

C		y_1		...	y_j	...	y_n	
		0	0	0	0	0	0	0
x_1	0							1
...	0							
x_i	0							
...	0							
x_m	0							
		0	1	...	j			

if $x_i = y_j$ **then**
 $LCS(X_i, Y_j) = LCS(X_{i-1}, Y_{j-1}) + \langle x_i \rangle$
else
if $C[i, j] = C[i-1, j]$ **then**
 $LCS(X_i, Y_j) = LCS(X_{i-1}, Y_j)$
else $// C[i, j] = C[i, j-1]$
 $LCS(X_i, Y_j) = LCS(X_i, Y_{j-1})$

[Ricostruzione di $\text{LCS}(X_i, Y_j)$]

C		y_1	...	y_j	...	y_n	
		0	0	0	0	0	0
x_1	0						1
...	0						...
x_i	0						i
...	0						
x_m	0						
		0	1	...	j	...	

$$\text{LCS}(X_i, Y_0) = \langle \rangle$$

]

$$\text{LCS}(X_i, Y_0) = \langle \rangle$$
$$\text{LCS}(X_0, Y_j) = \langle \rangle$$

[Ricostruzione di $\text{LCS}(X_i, Y_j)$]

```
procedura Print_LCS(C, X, i, j)
```

[Ricostruzione di $\text{LCS}(X_i, Y_j)$]

```
procedura Print_LCS(C, X, Y i, j)  
    if i > 0 and j > 0 then
```

[Ricostruzione di $\text{LCS}(X_i, Y_j)$]

```
procedura Print_LCS(C, X, Y i, j)
```

```
  if i > 0 and j > 0 then
```

```
    if  $x_i = y_j$  then
```

```
       $\text{LCS}(X_i, Y_j) = \text{LCS}(X_{i-1}, Y_{j-1}) + \langle x_i \rangle$ 
```

[Ricostruzione di $\text{LCS}(X_i, Y_j)$]

```
procedura Print_LCS(C, X, Y i, j)
  if i > 0 and j > 0 then
    if  $x_i = y_j$  then
      Print_LCS(C, X, i-1, j-1)
    print  $x_i$ 
```

[Ricostruzione di $\text{LCS}(X_i, Y_j)$]

```
procedura Print_LCS(C, X, Y i, j)
  if i > 0 and j > 0 then
    if  $x_i = y_j$  then
      Print_LCS(C, X, i-1, j-1)
      print  $x_i$ 
    else
      if  $C[i, j] = C[i-1, j]$  then
         $\text{LCS}(X_i, Y_j) = \text{LCS}(X_{i-1}, Y_j)$ 
```

[Ricostruzione di $\text{LCS}(X_i, Y_j)$]

```
procedura Print_LCS(C, X, Y i, j)
  if i > 0 and j > 0 then
    if  $x_i = y_j$  then
      Print_LCS(C, X, i-1, j-1)
      print  $x_i$ 
    else
      if C[i,j] = C[i-1,j] then
        Print_LCS(C, X, i-1, j)
```

[Ricostruzione di $\text{LCS}(X_i, Y_j)$]

```
procedura Print_LCS (C, X, Y i, j)
  if i > 0 and j > 0 then
    if  $x_i = y_j$  then
      Print_LCS (C, X, i-1, j-1)
      print  $x_i$ 
    else
      if  $C[i, j] = C[i-1, j]$  then
        Print_LCS (C, X, i-1, j)
      else //  $C[i, j]$  è uguale a  $C[i, j-1]$ 
         $\text{LCS}(X_i, Y_j) = \text{LCS}(X_i, Y_{j-1})$ 
```


[Ricostruzione di $\text{LCS}(X_i, Y_j)$]

```
procedura Print_LCS(C, X, Y i, j)
  if i > 0 and j > 0 then
    if  $x_i = y_j$  then
      Print_LCS(C, X, i-1, j-1)
      print  $x_i$ 
    else
      if  $C[i, j] = C[i-1, j]$  then
        Print_LCS(C, X, i-1, j)
      else //  $C[i, j]$  è uguale a  $C[i, j-1]$ 
        Print_LCS(C, X, i, j-1)
```



[Esercizio di ricostruzione

C		2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	1	2
5	0	1	2	2	2	2	2	2	2	3
3	0	1	2	2	2	3	3	3	3	4
1	0	1	2	2	2	3	3	4	4	5
12	0	1	2	3	3	3	4	4	4	6
	0	1	2	3	4	5	6	7	8	

$X = \langle 2, 10, 5, 3, 1, 12 \rangle$

$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$



Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = ?$$

C

2 5 12 2 3 12 1 30

	0	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	1	2
5	0	1	2	2	2	2	2	2	2	3
3	0	1	2	2	2	3	3	3	3	4
1	0	1	2	2	2	3	3	4	4	5
12	0	1	2	3	3	3	4	4	4	6
	0	1	2	3	4	5	6	7	8	

$$C_{6,8} = ?$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

]

Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = ?$$

C									
	2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	2
5	0	1	2	2	2	2	2	2	3
3	0	1	2	2	2	3	3	3	4
1	0	1	2	2	2	3	3	4	5
12	0	1	2	3	3	3	4	4	6
	0	1	2	3	4	5	6	7	8

$$C_{6,8} = C_{6,7} = C_{5,8}$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

]

Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = ?$$

C										
		2	5	12	2	3	12	1	30	
2 10 5 3 1	0	0	0	0	0	0	0	0	0	
	0	1	1	1	1	1	1	1	1	
	0	1	1	1	1	1	1	1	1	
	0	1	2	2	2	2	2	2	2	
	0	1	2	2	2	3	3	3	3	
	0	1	2	2	2	3	3	4	4	
12	0	1	2	3	3	3	4	4	4	
		0	1	2	3	4	5	6	7	8

$$C_{6,8} = \cancel{C_{6,7}} = C_{5,8}$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

]

Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_5, Y_8)$$

C										
		2	5	12	2	3	12	1	30	
2 10 5 3 1	0	0	0	0	0	0	0	0	0	
	0	1	1	1	1	1	1	1	1	
	0	1	1	1	1	1	1	1	1	
	0	1	2	2	2	2	2	2	2	
	0	1	2	2	2	3	3	3	3	
	0	1	2	2	2	3	3	4	4	
12	0	1	2	3	3	3	4	4	4	
		0	1	2	3	4	5	6	7	8

$$C_{6,8} = \cancel{C_{6,7}} = C_{5,8}$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$



Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_5, Y_8)$$

C

2 5 12 2 3 12 1 30

$\text{LCS}(X_5, Y_8) = ?$

	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	1
5	0	1	2	2	2	2	2	2	2
3	0	1	2	2	2	3	3	3	3
1	0	1	2	2	2	3	3	4	4
12	0	1	2	3	3	3	4	4	4
	0	1	2	3	4	5	6	7	8

$$C_{5,8} = ?$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

]

Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_5, Y_8)$$

C

2 5 12 2 3 12 1 30

$$\text{LCS}(X_5, Y_8) = \text{LCS}(X_5, Y_7)$$

	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	1
5	0	1	2	2	2	2	2	2	2
3	0	1	2	2	2	3	3	3	3
1	0	1	2	2	2	3	3	4	4
12	0	1	2	3	3	3	4	4	4
	0	1	2	3	4	5	6	7	8

$$C_{5,8} = C_{5,7}$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_5, Y_8)$$

C		2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	1	2
5	0	1	2	2	2	2	2	2	2	3
3	0	1	2	2	2	3	3	3	3	4
1	0	1	2	2	2	3	3	4	4	5
12	0	1	2	3	3	3	4	4	4	6
	0	1	2	3	4	5	6	7	8	

$$\text{LCS}(X_5, Y_8) = \text{LCS}(X_5, Y_7)$$

$$C_{5,8} = C_{5,7}$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

]

Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_5, Y_7)$$

C									
	2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	2
5	0	1	2	2	2	2	2	2	3
3	0	1	2	2	2	3	3	3	4
1	0	1	2	2	2	3	3	4	5
12	0	1	2	3	3	3	4	4	6
	0	1	2	3	4	5	6	7	8

$$C_{5,8} = C_{5,7}$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

[Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_5, Y_7)$$

C

2 5 12 2 3 12 1 30

2

10

5

3

1

12

0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	
0	1	1	1	1	1	1	1	1	
0	1	2	2	2	2	2	2	2	
0	1	2	2	2	3	3	3	3	
0	1	2	2	2	3	3	4	4	
0	1	2	3	3	3	4	4	4	

$\text{LCS}(X_5, Y_7) = ?$

$C_{5,7} = ?$

$X = \langle 2, 10, 5, 3, 1, 12 \rangle$

$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$

[Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_5, Y_7)$$

C

2 5 12 2 3 12 1 30

	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	
10	0	1	1	1	1	1	1	1	2
5	0	1	2	2	2	2	2	2	3
3	0	1	2	2	2	3	3	3	4
1	0	1	2	2	2	3	3	4	5
12	0	1	2	3	3	3	4	4	6
	0	1	2	3	4	5	6	7	8

$$\text{LCS}(X_5, Y_7) = \text{LCS}(X_4, Y_6) + \langle x_5 \rangle$$

$$C_{5,7} = C_{4,6} + 1$$

$X = \langle 2, 10, 5, 3, 1, 12 \rangle$

$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$

[Esercizio di ricostruzione]

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_5, Y_7)$$

C		2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	LCS(X	
10	0	1	1	1	1	1	1	1	1	2
5	0	1	2	2	2	2	2	2	2	3
3	0	1	2	2	2	3	3	3	3	4
1	0	1	2	2	2	3	3	4	4	5
12	0	1	2	3	3	3	4	4	4	6
	0	1	2	3	4	5	6	7	8	

$$\text{LCS}(X_5, Y_7) = \text{LCS}(X_4, Y_6) + \langle x_5 \rangle$$

$$c_{5,7} = c_{4,6} + 1$$

$X = \langle 2, 10, 5, 3, 1, 12 \rangle$

$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$

[Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_4, Y_6) + \langle x_5 \rangle$$

C									
		2	5	12	2	3	12	1	30
	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	1
5	0	1	2	2	2	2	2	2	2
3	0	1	2	2	2	3	3	3	3
1	0	1	2	2	2	3	3	4	4
12	0	1	2	3	3	3	4	4	4
	0	1	2	3	4	5	6	7	8

$$C_{5,7} = C_{4,6} + 1$$

$X = \langle 2, 10, 5, 3, 1, 12 \rangle$

$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$



Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_4, Y_6) + \langle x_5 \rangle$$

$$\text{LCS}(X_4, Y_6) = ?$$

C

2 5 12 2 3 12 1 30

	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	1
5	0	1	2	2	2	2	2	2	2
3	0	1	2	2	2	3	3	3	3
1	0	1	2	2	2	3	3	4	4
12	0	1	2	3	3	3	4	4	4
	0	1	2	3	4	5	6	7	8

$$C_{4,6} = ?$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

]

Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_4, Y_6) + \langle x_5 \rangle$$

$$\text{LCS}(X_4, Y_6) = \text{LCS}(X_4, Y_5)$$

C

		2	5	12	2	3	12	1	30
0	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	1
5	0	1	2	2	2	2	2	2	2
3	0	1	2	2	2	3	3	3	3
1	0	1	2	2	2	3	3	4	4
12	0	1	2	3	3	3	4	4	4
0	0	1	2	3	4	5	6	7	8

	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	1
5	0	1	2	2	2	2	2	2	2
3	0	1	2	2	2	3	3	3	3
1	0	1	2	2	2	3	3	4	4
12	0	1	2	3	3	3	4	4	4
	0	1	2	3	4	5	6	7	8

$$C_{4,6} = C_{4,5}$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

[Esercizio di ricostruzione]

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_4, Y_6) + \langle x_5 \rangle$$

$$\text{LCS}(X_4, Y_6) = \text{LCS}(X_4, Y_5)$$

C

2 5 12 2 3 12 1 30

	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	1
5	0	1	2	2	2	2	2	2	2
3	0	1	2	2	2	3	3	3	3
1	0	1	2	2	2	3	3	4	4
12	0	1	2	3	3	3	4	4	4

0 1 2 3 4 5 6 7 8

$$C_{4,6} = C_{4,5}$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

]

Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_4, Y_5) + \langle x_5 \rangle$$

C									
		2	5	12	2	3	12	1	30
0	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	1
5	0	1	2	2	2	2	2	2	2
3	0	1	2	2	2	3	3	3	3
1	0	1	2	2	2	3	3	4	4
12	0	1	2	3	3	3	4	4	4
	0	1	2	3	4	5	6	7	8

$$C_{4,6} = C_{4,5}$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$



Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_4, Y_5) + \langle x_5 \rangle$$

C

2 5 12 2 3 12 1 30

	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1		
10	0	1	1	1	1	1	1	1	2
5	0	1	2	2	2	2	2	2	3
3	0	1	2	2	2	3	3	3	4
1	0	1	2	2	2	3	3	4	5
12	0	1	2	3	3	3	4	4	6
	0	1	2	3	4	5	6	7	8

$\text{LCS}(X_4, Y_5) = ?$

$C_{4,5} = ?$

$X = \langle 2, 10, 5, 3, 1, 12 \rangle$

$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$

[Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_4, Y_5) + \langle x_5 \rangle$$

C

2 5 12 2 3 12 1 30

	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	
10	0	1	1	1	1	1	1	1	2
5	0	1	2	2	2	2	2	2	3
3	0	1	2	2	2	3	3	3	4
1	0	1	2	2	2	3	4	4	5
12	0	1	2	3	3	3	4	4	6
	0	1	2	3	4	5	6	7	8

$$\text{LCS}(X_4, Y_5) = \text{LCS}(X_3, Y_4) + \langle x_4 \rangle$$

$$c_{4,5} = c_{3,4} + 1$$

$X = \langle 2, 10, 5, 3, 1, 12 \rangle$

$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$

]

Esercizio di ricostruzione

$$LCS(X_6, Y_8) = LCS(X_4, Y_5) + \langle x_5 \rangle$$

C

2 5 12 2 3 12 1 30

2

10

5

3

1

12

0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	
0	1	1	1	1	1	1	1	1	2
0	1	2	2	2	2	2	2	2	3
0	1	2	2	2	3	3	3	3	4
0	1	2	2	2	3	3	4	4	5
0	1	2	3	3	3	4	4	4	6

0 1 2 3 4 5 6 7 8

$$LCS(X_4, Y_5) = LCS(X_3, Y_4) + \langle x_4 \rangle$$

$$C_{4,5} = C_{3,4} + 1$$

X = <2, 10, 5, 3, 1, 12>

Y = <2, 5, 12, 2, 3, 12, 1, 30>

]

Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_3, Y_4) + \langle x_4 \rangle + \langle x_5 \rangle$$

C

2 5 12 2 3 12 1 30

	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	2
5	0	1	2	2	2	2	2	2	3
3	0	1	2	2	2	3	3	3	4
1	0	1	2	2	2	3	3	4	5
12	0	1	2	3	3	3	4	4	6
	0	1	2	3	4	5	6	7	8

$$c_{4,5} = c_{3,4} + 1$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

]

Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_3, Y_4) + \langle x_4 \rangle + \langle x_5 \rangle$$

C

2 5 12 2 3 12 1 30

	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	1
5	0	1	2	2	2	2	2	2	2
3	0	1	2	2	2	3	3	3	3
1	0	1	2	2	2	3	3	4	4
12	0	1	2	3	3	3	4	4	4
	0	1	2	3	4	5	6	7	8

$$\text{LCS}(X_3, Y_4) = ?$$

$$C_{3,4} = ?$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

]

Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_3, Y_4) + \langle x_4 \rangle + \langle x_5 \rangle$$

C

2 5 12 2 3 12 1 30

0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1
0	1	1	1	1	1	1	1	1
0	1	2	2	2	2	2	2	2
0	1	2	2	2	3	3	3	3
0	1	2	2	2	3	4	4	4
0	1	2	3	3	3	4	4	4
0	1	2	3	4	5	6	7	8

$$\text{LCS}(X_3, Y_4) = \text{LCS}(X_3, Y_3)$$

$$C_{3,4} = C_{3,3}$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

]

Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_3, Y_4) + \langle x_4 \rangle + \langle x_5 \rangle$$

C

2 5 12 2 3 12 1 30

	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	1
5	0	1	2	2	2	2	2	2	2
3	0	1	2	2	2	3	3	3	3
1	0	1	2	2	2	3	3	4	4
12	0	1	2	3	3	3	4	4	4
	0	1	2	3	4	5	6	7	8

$$\text{LCS}(X_3, Y_4) = \text{LCS}(X_3, Y_3)$$

$$C_{3,4} = C_{3,3}$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

]

Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_3, Y_3) + \langle x_4 \rangle + \langle x_5 \rangle$$

C

2 5 12 2 3 12 1 30

	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	1
5	0	1	2	2	2	2	2	2	2
3	0	1	2	2	2	3	3	3	3
1	0	1	2	2	2	3	3	4	4
12	0	1	2	3	3	3	4	4	4
	0	1	2	3	4	5	6	7	8

$$C_{3,4} = C_{3,3}$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$



Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_3, Y_3) + \langle x_4 \rangle + \langle x_5 \rangle$$

C

2 5 12 2 3 12 1 30

2

10

5

3

1

12

0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1
0	1	1	1	1	1	1	1	1
0	1	2	2	2	2	2	2	2
0	1	2	2	2	3	3	3	3
0	1	2	2	2	3	3	4	4
0	1	2	3	3	3	4	4	4
0	1	2	3	4	5	6	7	8

$\text{LCS}(X_3, Y_3) = ?$

$C_{3,3} = ?$

$X = \langle 2, 10, 5, 3, 1, 12 \rangle$

$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$

]

Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_3, Y_3) + \langle x_4 \rangle + \langle x_5 \rangle$$

C

2 5 12 2 3 12 1 30

$$\text{LCS}(X_3, Y_3) = \text{LCS}(X_3, Y_2)$$

2

10

5

3

1

12

	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	1	1	1
1	0	1	1	1	1	1	1	1	1
2	0	1	2	2	2	2	2	2	2
3	0	1	2	2	2	3	3	3	3
4	0	1	2	2	2	3	3	4	4
5	0	1	2	3	3	3	4	4	4
6	0	1	2	3	3	3	4	4	4
7	0	1	2	3	3	3	4	4	4
8	0	1	2	3	3	3	4	4	4

$$C_{3,3} = C_{3,2}$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

]

Esercizio di ricostruzione

$$LCS(X_6, Y_8) = LCS(X_3, Y_3) + \langle x_4 \rangle + \langle x_5 \rangle$$

C

2 5 12 2 3 12 1 30

	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	1
5	0	1	2	2	2	2	2	2	2
3	0	1	2	2	2	3	3	3	3
1	0	1	2	2	2	3	3	4	4
12	0	1	2	3	3	3	4	4	4

0 1 2 3 4 5 6 7 8

$$LCS(X_3, Y_3) = LCS(X_3, Y_2)$$

$$C_{3,3} = C_{3,2}$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

[Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_3, Y_2) + \langle x_4 \rangle + \langle x_5 \rangle$$

C		2	5	12	2	3	12	1	30	
		0	0	0	0	0	0	0	0	0
2	0	0	1	1	1	1	1	1	1	1
10	0	0	1	1	1	1	1	1	1	2
5	0	0	1	2	2	2	2	2	2	3
3	0	0	1	2	2	2	3	3	3	4
1	0	0	1	2	2	2	3	3	4	5
12	0	0	1	2	3	3	3	4	4	6
	0	1	2	3	4	5	6	7	8	

$$C_{3,3} = C_{3,2}$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

[Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_3, Y_2) + \langle x_4 \rangle + \langle x_5 \rangle$$

C 2 5 12 2 3 12 1 30

	0	0	0	0	0	0	0	0	0	
2	0	1	1	1	1	1	1	1		
10	0	1	1	1	1	1	1	1	2	
5	0	1	2	2	2	2	2	2	3	
3	0	1	2	2	2	3	3	3	4	
1	0	1	2	2	2	3	3	4	5	
12	0	1	2	3	3	3	4	4	6	
	0	1	2	3	4	5	6	7	8	

$\text{LCS}(X_3, Y_2) = ?$

$C_{3,2} = ?$

$X = \langle 2, 10, 5, 3, 1, 12 \rangle$

$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$

]

Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_3, Y_2) + \langle x_4 \rangle + \langle x_5 \rangle$$

C

2

5

12

2

3

12

1

30

0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	LCS(X	
0	1	1	1	1	1	1	1	1	2
0	1	2	2	2	2	2	2	2	3
0	1	2	2	2	3	3	3	3	4
0	1	2	2	2	3	3	4	4	5
0	1	2	3	3	3	4	4	4	6
0	1	2	3	4	5	6	7	8	

$$\text{LCS}(X_3, Y_2) = \text{LCS}(X_2, Y_1) + \langle x_3 \rangle$$

$$C_{3,2} = C_{2,1} + 1$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

]

Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_3, Y_2) + \langle x_4 \rangle + \langle x_5 \rangle$$

C

2 5 12 2 3 12 1 30

	0	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1		
10	0	1	1	1	1	1	1	1	2	
5	0	1	2	2	2	2	2	2	3	
3	0	1	2	2	2	3	3	3	4	
1	0	1	2	2	2	3	3	4	5	
12	0	1	2	3	3	3	4	4	6	
	0	1	2	3	4	5	6	7	8	

$$\text{LCS}(X_3, Y_2) = \text{LCS}(X_2, Y_1) + \langle x_3 \rangle$$

$$C_{3,2} = C_{2,1} + 1$$

$X = \langle 2, 10, 5, 3, 1, 12 \rangle$

$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$

[Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_2, Y_1) + \langle x_3 \rangle + \langle x_4 \rangle + \langle x_5 \rangle$$

C

2 5 12 2 3 12 1 30

	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	1
5	0	1	2	2	2	2	2	2	2
3	0	1	2	2	2	3	3	3	3
1	0	1	2	2	2	3	3	4	4
12	0	1	2	3	3	3	4	4	4
	0	1	2	3	4	5	6	7	8

$$C_{3,2} = C_{2,1} + 1$$

$X = \langle 2, 10, 5, 3, 1, 12 \rangle$

$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$

[Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_2, Y_1) + \langle x_3 \rangle + \langle x_4 \rangle + \langle x_5 \rangle$$

C		2	5	12	2	3	12	1	30	
		0	0	0	0	0	0	0	0	0
2	0	0	1	1	1	1	1	1	1	LCS(X ₂ , Y ₁) = ?
10	0	0	1	1	1	1	1	1	1	
5	0	0	1	2	2	2	2	2	2	3
3	0	0	1	2	2	2	3	3	3	4
1	0	0	1	2	2	2	3	4	4	5
12	0	0	1	2	3	3	3	4	4	6
	0	1	2	3	4	5	6	7	8	

$$C_{2,1} = ?$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

[Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_2, Y_1) + \langle x_3 \rangle + \langle x_4 \rangle + \langle x_5 \rangle$$

C

2

5

12

2

3

12

1

30

	0	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1		
10	0	1	1	1	1	1	1	1	1	2
5	0	1	2	2	2	2	2	2	2	3
3	0	1	2	2	2	3	3	3	3	4
1	0	1	2	2	2	3	3	4	4	5
12	0	1	2	3	3	3	4	4	4	6
	0	1	2	3	4	5	6	7	8	

$$\text{LCS}(X_2, Y_1) = \text{LCS}(X_1, Y_1)$$

$$C_{2,1} = C_{1,1}$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

[Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_2, Y_1) + \langle x_3 \rangle + \langle x_4 \rangle + \langle x_5 \rangle$$

C

2 5 12 2 3 12 1 30

	0	0	0	0	0	0	0	0	0	
2	0	1	1	1	1	1	1	1		
10	0	1	1	1	1	1	1	1	1	2
5	0	1	2	2	2	2	2	2	2	3
3	0	1	2	2	2	3	3	3	3	4
1	0	1	2	2	2	3	3	4	4	5
12	0	1	2	3	3	3	4	4	4	6
	0	1	2	3	4	5	6	7	8	

$$\text{LCS}(X_2, Y_1) = \text{LCS}(X_1, Y_1)$$

$$C_{2,1} = C_{1,1}$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

[Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_1, Y_1) + \langle x_3 \rangle + \langle x_4 \rangle + \langle x_5 \rangle$$

C		2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	1	2
5	0	1	2	2	2	2	2	2	2	3
3	0	1	2	2	2	3	3	3	3	4
1	0	1	2	2	2	3	3	4	4	5
12	0	1	2	3	3	3	4	4	4	6
	0	1	2	3	4	5	6	7	8	

$$C_{2,1} = C_{1,1}$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

[Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_1, Y_1) + \langle x_3 \rangle + \langle x_4 \rangle + \langle x_5 \rangle$$

C		2	5	12	2	3	12	1	30	
		0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1		
10	0	1	1	1	1	1	1	1	2	
5	0	1	2	2	2	2	2	2	3	
3	0	1	2	2	2	3	3	3	4	
1	0	1	2	2	2	3	3	4	5	
12	0	1	2	3	3	3	4	4	6	
	0	1	2	3	4	5	6	7	8	

$\text{LCS}(X_1, Y_1) = ?$

$C_{1,1} = ?$

$X = \langle 2, 10, 5, 3, 1, 12 \rangle$

$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$

[Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_1, Y_1) + \langle x_3 \rangle + \langle x_4 \rangle + \langle x_5 \rangle$$

C

2

5

12

2

3

12

1

30

	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	
10	0	1	1	1	1	1	1	1	2
5	0	1	2	2	2	2	2	2	3
3	0	1	2	2	2	3	3	3	4
1	0	1	2	2	2	3	4	4	5
12	0	1	2	3	3	3	4	4	6
	0	1	2	3	4	5	6	7	8

$$\text{LCS}(X_1, Y_1) = \text{LCS}(X_0, Y_0) + \langle x_1 \rangle$$

$$c_{1,1} = c_{0,0} + 1$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

]

Esercizio di ricostruzione

$$LCS(X_6, Y_8) = LCS(X_1, Y_1) + \langle x_3 \rangle + \langle x_4 \rangle + \langle x_5 \rangle$$

C

2

5

12

2

3

12

1

30

0

0

0

0

0

0

0

0

0

0

2

0

1

1

1

1

1

1

1

$$LCS(X_1, Y_1) = LCS(X_0, Y_0) + \langle x_1 \rangle$$

10

0

1

1

1

1

1

1

1

1

2

5

0

1

2

2

2

2

2

2

2

3

3

0

1

2

2

2

3

3

3

3

4

1

0

1

2

2

2

3

3

4

4

5

12

0

1

2

3

3

3

4

4

4

6

0

1

2

3

4

5

6

7

8

$$C_{1,1} = C_{0,0} + 1$$

$X = \langle 2, 10, 5, 3, 1, 12 \rangle$

$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$

Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_0, Y_0) + \langle x_1 \rangle + \langle x_3 \rangle + \langle x_4 \rangle + \langle x_5 \rangle$$

C		2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0	0
	2	0	1	1	1	1	1	1	1	1
	10	0	1	1	1	1	1	1	1	2
	5	0	1	2	2	2	2	2	2	3
	3	0	1	2	2	2	3	3	3	4
	1	0	1	2	2	2	3	3	4	5
	12	0	1	2	3	3	3	4	4	6
	0	1	2	3	4	5	6	7	8	

$$c_{1,1} = c_{0,0} + 1$$

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

[Esercizio di ricostruzione

$$LCS(X_6, Y_8) = LCS(X_0, Y_0) + \langle x_1 \rangle + \langle x_3 \rangle + \langle x_4 \rangle + \langle x_5 \rangle$$

C **2 5 12 2 3 12 1 30**

	0	0	0	0	0	0	0	0	0	
2	0	1	1	1	1	1	1	1		
10	0	1	1	1	1	1	1	1	2	
5	0	1	2	2	2	2	2	2	3	
3	0	1	2	2	2	3	3	3	4	
1	0	1	2	2	2	3	3	4	5	
12	0	1	2	3	3	3	4	4	6	
	0	1	2	3	4	5	6	7	8	

$LCS(X_0, Y_0) = ?$

$$C_{0,0} = 0$$

$X = \langle \mathbf{2}, \mathbf{10}, \mathbf{5}, \mathbf{3}, \mathbf{1}, \mathbf{12} \rangle$

$Y = \langle \mathbf{2}, \mathbf{5}, \mathbf{12}, \mathbf{2}, \mathbf{3}, \mathbf{12}, \mathbf{1}, \mathbf{30} \rangle$



Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \text{LCS}(X_0, Y_0) + \langle x_1 \rangle + \langle x_3 \rangle + \langle x_4 \rangle + \langle x_5 \rangle$$

C 2 5 12 2 3 12 1 30

	0	0	0	0	0	0	0	0	0	
2	0	1	1	1	1	1	1	1		
10	0	1	1	1	1	1	1	1	2	
5	0	1	2	2	2	2	2	2	3	
3	0	1	2	2	2	3	3	3	4	
1	0	1	2	2	2	3	3	4	5	
12	0	1	2	3	3	3	4	4	6	
	0	1	2	3	4	5	6	7	8	

$\text{LCS}(X_0, Y_0) = \langle \rangle$

$$c_{0,0} = 0$$

$X = \langle 2, 10, 5, 3, 1, 12 \rangle$

$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$

]

Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \langle \rangle + \langle x_1 \rangle + \langle x_3 \rangle + \langle x_4 \rangle + \langle x_5 \rangle$$

C 2 5 12 2 3 12 1 30

	0	0	0	0	0	0	0	0	0	
2	0	1	1	1	1	1	1	1		
10	0	1	1	1	1	1	1	1	2	
5	0	1	2	2	2	2	2	2	3	
3	0	1	2	2	2	3	3	3	4	
1	0	1	2	2	2	3	3	4	5	
12	0	1	2	3	3	3	4	4	6	
	0	1	2	3	4	5	6	7	8	

$\text{LCS}(X_0, Y_0) = \langle \rangle$

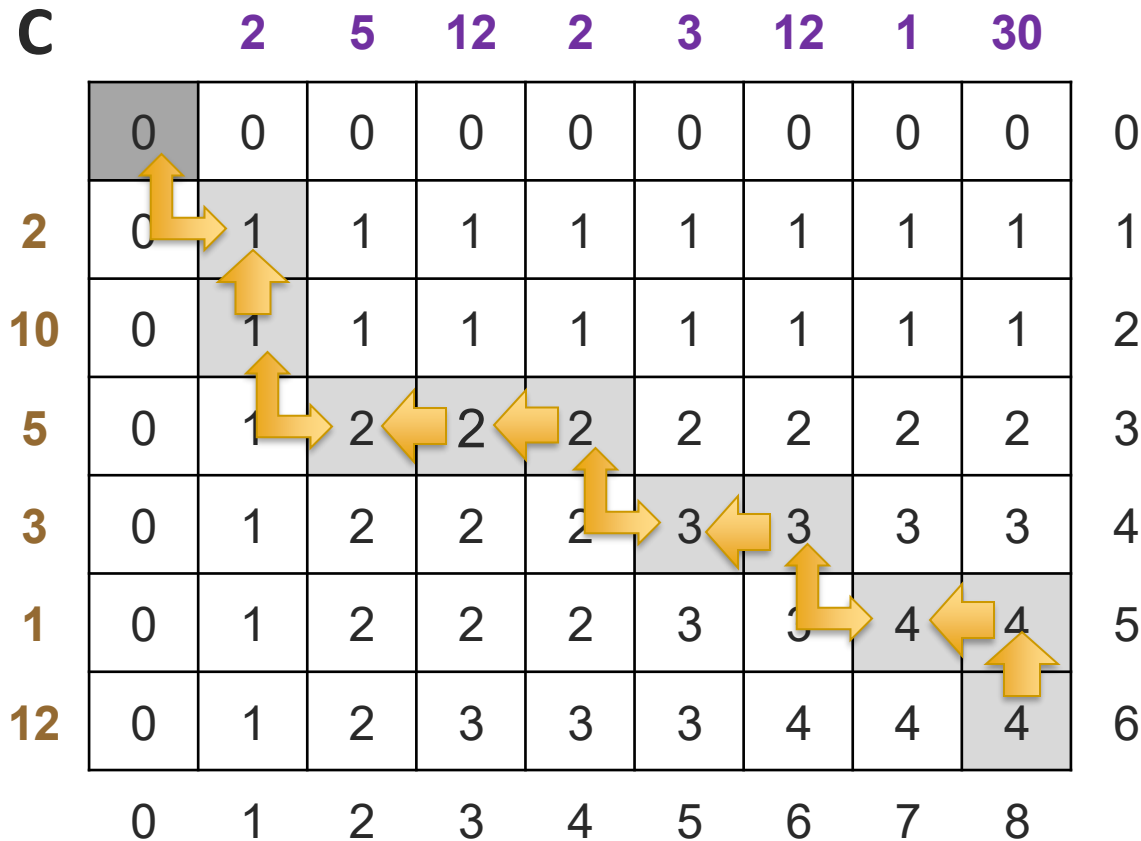
$$c_{0,0} = 0$$

$X = \langle 2, 10, 5, 3, 1, 12 \rangle$

$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$

[Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \langle x_1, x_3, x_4, x_5 \rangle$$



$X = \langle 2, 10, 5, 3, 1, 12 \rangle$

$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$

[Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \langle 2, 5, 3, 1 \rangle$$

C

		2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	1	2
5	0	1	2	2	2	2	2	2	2	3
3	0	1	2	2	2	3	3	3	3	4
1	0	1	2	2	2	3	3	4	4	5
12	0	1	2	3	3	3	4	4	4	6
	0	1	2	3	4	5	6	7	8	

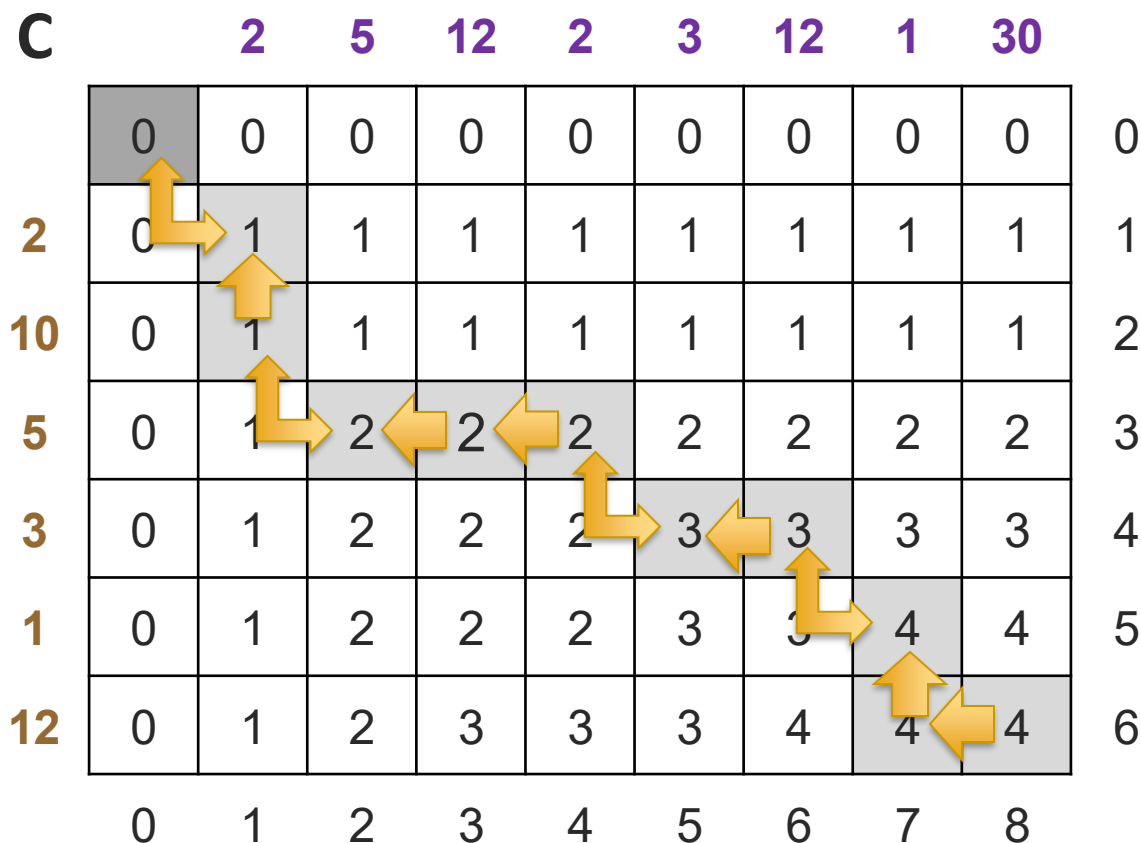
$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$



Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \langle 2, 5, 3, 1 \rangle$$



Percorso alternativo che
non porta a una diversa
soluzione

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

Esercizio di ricostruzione

$$\text{LCS}(X_6, Y_8) = \langle 2, 5, 3, 12 \rangle$$

C

		2	5	12	2	3	12	1	30	
	0	0	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	1	1	1
10	0	1	1	1	1	1	1	1	1	2
5	0	1	2	2	2	2	2	2	2	3
3	0	1	2	2	2	3	3	3	3	4
1	0	1	2	2	2	3	3	4	4	5
12	0	1	2	3	3	3	4	4	4	6
	0	1	2	3	4	5	6	7	8	

Percorso alternativo che porta a una diversa soluzione

$$X = \langle 2, 10, 5, 3, 1, 12 \rangle$$

$$Y = \langle 2, 5, 12, 2, 3, 12, 1, 30 \rangle$$

[Hardest Common Subsequence (HCS)]

Given two sequences

$$X = \langle x_1, x_2, \dots, x_m \rangle, Y = \langle y_1, y_2, \dots, y_n \rangle$$

and a weight function $w : \Sigma \rightarrow \mathbb{N}$, find the most heavy common subsequence $Z = \langle z_1, z_2, \dots, z_k \rangle$, i.e. the one for which the total weight:

$$W(Z) = \sum_{i=1}^k w(z_i)$$

is maximum

[Sottoproblema]

Sottoproblema di dimensione (i,j)

→ Trovare $HCS(X_i, Y_j)$

$i \in \{0, 1, \dots, m\}$

$j \in \{0, 1, \dots, n\}$

Numero totale di sottoproblemi: $(m+1) \times (n+1)$

[Sottoproblema]

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$$c_{i,j} = W(HCS(X_i, Y_j))$$

[Sottoproblema]

Sottoproblema di dimensione (i,j)

→ Trovare $HCS(X_i, Y_j)$

$i \in \{0, 1, \dots, m\}$

$j \in \{0, 1, \dots, n\}$

Numero totale di sottoproblemi: $(m+1) \times (n+1)$

$$c_{m,n} = W(HCS(X_m, Y_n)) = W(HCS(X, Y))$$

[Equazioni di ricorrenza]

$i = 0 \vee j = 0$ (CASI BASE)

$HCS(X_i, Y_j) =$

[Equazioni di ricorrenza]

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$$X_i = \langle x_1, x_2, \dots, x_{i-1}, x_i \rangle$$

$$Y_j = \langle y_1, y_2, \dots, y_{j-1}, y_j \rangle$$

$i > 0 \wedge j > 0$ (PASSO RICORSIVO)

$$x_i = y_j \Rightarrow \text{HCS}(X_i, Y_j) =$$

$$x_i \neq y_j \Rightarrow \text{HCS}(X_i, Y_j) =$$

[Equazioni di ricorrenza]

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$i > 0 \wedge j > 0$ (PASSO RICORSIVO)

$$x_i = y_j \Rightarrow \text{HCS}(X_i, Y_j) = \text{HCS}(X_{i-1}, Y_{j-1}) + \langle x_i \rangle$$

$$x_i \neq y_j \Rightarrow \text{HCS}(X_i, Y_j) = \max\{\text{HCS}(X_{i-1}, Y_j), \text{HCS}(X_i, Y_{j-1})\}$$

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[Equazioni di ricorrenza]

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$$\Rightarrow c_{i,j} = c_{i-1,j-1} + w(x_i)$$

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[Equazioni di ricorrenza]

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$$X_i = \langle x_1, x_2, \dots, x_{i-1}, x_i \rangle$$

$$Y_j = \langle y_1, y_2, \dots, y_{j-1}, y_j \rangle$$

$i > 0 \wedge j > 0$ (PASSO RICORSIVO)

$$x_i = y_j \Rightarrow c_{i,j} = c_{i-1,j-1} + w(x_i)$$

$$x_i \neq y_j \Rightarrow c_{i,j} = \max\{c_{i-1,j}, c_{i,j-1}\}$$

[Soluzione del problema (DP)]

1. Calcolo del valore ottimo $c_{m,n}$ con un algoritmo bottom-up
 - memorizzazione dei coefficienti
2. Ricostruzione di $HCS(X, Y)$